7

Triangles

A truss bridge is a bridge whose load-bearing superstructure is made up of trusses, which are connected elements that usually form triangular units. On both sides of a truss bridge, equilateral triangles are formed and all of these triangles are congruent under the SSS criteria. This is because the truss bridge requires equal weight controlling lengths to keep the structure intact and prevent it from collapsing.

What's Inside

Topic(s)

- ✓ Topic-wise Notes
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- ✓ Caution Points
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Questions

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TOPIC 1

CONGRUENCE OF TRIANGLES

Learning Objectives

- Students will be learning the different types of triangles and their properties.
- Students will be learning the concept of congruency of triangles and its axioms.

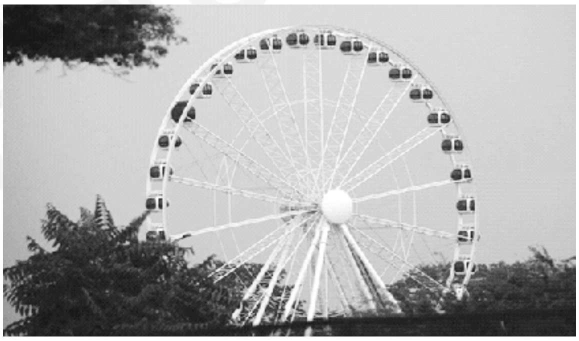
Learning Outcomes

- Students will be able to understand different congruence conditions of triangles.
- Students will be able to develop higher order thinking skills by identifying various criteria for congruency of triangles for various figures.

Real Life Application

You must have heard about Delhi's eye. It is India's tallest Ferris wheel and the fifth tallest in the world which is around 250 ft high. It provides a bird's-eye view of Delhi, Noida, Faridabad, and adjoining areas.

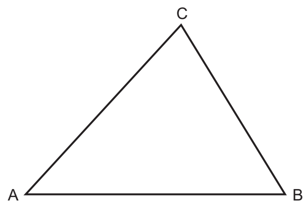
A Ferris wheel is an amusement ride which consists of a rotating upright wheel with multiple passenger-carrying components attached to the rim in such a way that as the wheel turns, they stand straight.



Have you ever imagined how this Ferris wheel was designed and built? Congruence of triangles are used by engineers and architects to build Ferris wheel. The Ferris wheel's spokes are made up of triangles that are congruent to each other which means that they all are of the same size and shape under the SAS criteria.

The concept of wheel congruence aids in achieving balance during ride. Congruent triangles are employed in the building process to strengthen the framework. This is how the congruence of triangles applies in our day *to day life. Let us now examine and comprehend all the criteria for triangle congruence.*

A closed figure formed by three intersecting lines is called a triangle. It has three sides, three angles and three vertices as shown in the figure.



In triangle ABC (denoted as ΔABC), AB, BC and CA are the three sides; ∠A, ∠B and ∠C are the three angles; and A, B and C are the three vertices.

Important

➥ *A triangle is the smallest polygon having three sides.*

If two figures are of the same shape and also of the same size, *i.e.*, superimpose to each other then they will coincide throughout thus, they are said to be congruent figures.

For example, wheels of a bicycle, pages of a textbook, two mobile phones of the same brand and same model, etc.

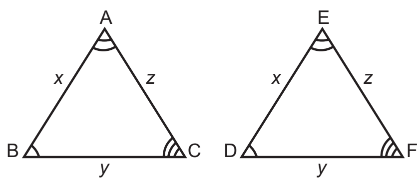
Important

➥ *Congruent triangles mean equal in all respects, or figures whose shape and size both are same.*

In geometry, two geometric figures are said to be congruent, if it is possible to superimpose one of them on the other so that they coincide throughout.

For example, two circles of the same radius, two squares of the same side, etc. If they are placed one on the other, then they will superimpose each other completely.

In the same way, two triangles are congruent if and only if one of them can be made to superimpose on the other, to cover it exactly. In other words, two triangles are said to be congruent if all the sides of a triangle are equal to all the corresponding sides of another triangle.



In the above figure, AB covers ED, BC covers DF and CA covers FE; ∠A covers ∠E, ∠B covers ∠D and ∠C covers ∠F. Also, there is a one-one correspondence between the vertices, *i.e.*, A corresponds to E (A ↔ E), B to D (B ↔ D), and C to F (C ↔ F).

Hence, under this correspondence we can say, ΔABC is congruent to triangle ΔEDF and it is represented as

$$\Delta ABC \cong \Delta EDF.$$

From the above definition, it is clear that AB = ED, BC = DF, CA = FE and $\angle A = \angle E$, $\angle B = \angle D$ and $\angle C = \angle F$.

Caution

➥ *$\Delta ABC \cong \Delta EDF$ is not same as $\Delta ABC \cong \Delta DEF$ or $\Delta ABC \cong \Delta DFE$, as A corresponds to E ($A \leftrightarrow E$), B to D ($B \leftrightarrow D$), and C to F ($C \leftrightarrow F$).*

Important

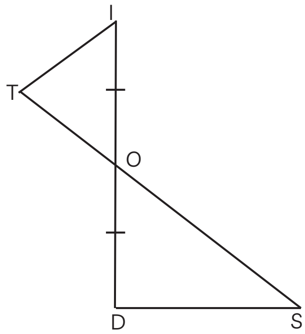
➥ *In congruent triangles, corresponding parts are equal and we write in short 'CPCT' (Corresponding parts of congruent triangles).*

OBJECTIVE Type Questions

[**1** mark]

Multiple Choice Questions

1. The triangles made by two intersecting lines as shown in figure. What additional information is required to prove that $\Delta TOI \cong \Delta SOD$?



(a) $\angle DOS = \angle TOI$ **(b) $\angle OTI = \angle ODI$**
(c) $TO = OS$ **(d) $TI = DS$** **[Delhi Gov. SQP 2022]**

Ans.(*b*) $\angle OTI = \angle ODI$

Explanation: According to the SAS Congruence criteria, two triangles are congruent if two sides and the included angle of one triangle are equal to the two sides and the included angle of the other triangle.

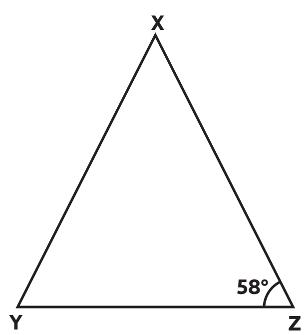
So, In ΔTOI and ΔSOD,

$OI = OD$ ..(i)
[sides of the triangle]

$\angle OTI = \angle ODI$ ..(ii)
[included angles of the triangle]

$OT = DI$
[sides of the triangle so they are equal]

2. ΔXYZ is an isosceles triangle, such that $XY = XZ$ and $\angle Z = 58°$. The value of $\angle X + \angle Y$ is:



(a) 180° (b) 64°
(c) 122° (d) 116° **[Diksha]**

Ans. *(c) 122°*

Explanation: In ΔXYZ is an isosceles triangle.

$XY = XZ$ [Given]

$\angle Z = \angle Y$

[Angles opposite to equal sides are equal]

$\therefore \quad \angle Y = \angle 58°$ ...(i)

$\angle X + \angle Y + \angle Z = 180°$

[Angle sum property of triangles]

$\angle X + 58° + 58° = 180°$

$\angle X + 116° = 180°$

$\angle X = 64°$ ...(ii)

Adding eqs. (i) and (ii)

$\angle X + \angle Y = 64° + 58°$

$\angle X + \angle Y = 122°$

True and False

3. **Equality of three angles is not sufficient for the congruence of triangles.**

Ans. *True*

Explanation: Equality of three angles is not sufficient for the congruence of triangles. Therefore, for the congruence of triangles out of three equal parts, one has to be a side.

Assertion and Reason (A-R)

In these questions, a statement of Assertion (A) is followed by a statement of Reason (R).

Choose the correct option as:

(a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).

(b) Both assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A).

(c) Assertion (A) is true but reason (R) is false.

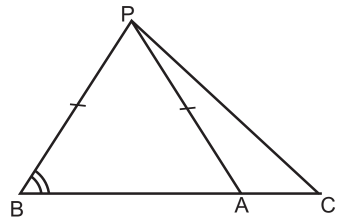
(d) Assertion (A) is false but reason (R) is true.

4. **Assertion (A): In the figure, A is a point on side BC of ΔPBC such that PB = PA, then PC > PB.**

Reason (R):In a triangle, the side opposite to the larger angle is longer.

Ans.(a) *Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).*

Explanation:



In ΔPBA,

$PB = PA$ [Given]

So, $\angle PBA = \angle PAB = x$

[Opposite angle of equal sides are equal]

$\angle PAC = \angle PBA + \angle PAB$

[Exterior angle property]

$\angle PAC = x + x$

$\angle PAC = 2x$

$\angle PAC > \angle PAB$

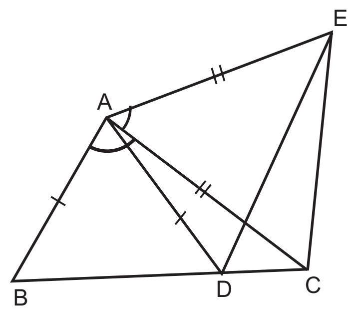
$\therefore PC > PB$ [Side opposite to larger angle]

CASE BASED Questions (CBQs)

[**4** & **5** marks]

Read the following passages and answer the questions that follow:

5. *Ramanuj wants to grow potatoes, so he bought farmland in quadrilateral form. But later on, he decided to grow two types of potatoes and the same mix of both the types so he made several divisions on the land as shown in the figure. Here, AC = AE, AB = AD and ∠BAC = ∠EAC.*



(A)$\Delta ABC \cong \Delta ADE$ by which congruence rule?

(a) AAS (b) SAS

(c) ASA (d) AAA

(B) Which among the following conditions is required to make $\Delta ABC \cong \Delta ADE$ by SAS rule, if only AB = AD and AC = AE are given?

(a) $\angle ABC = \angle AED$

(b) $\angle CAB = \angle EDA$

(c) $\angle BAC = \angle DAE$

(d) None of these

(C) If $\Delta ABC \cong \Delta ADE$, then BC is equal to which among the following sides?

(a) AB (b) EC

(c) AD (d) DE

(D) If $\angle BAD + \angle DAC = x + \angle DAC$, then find the value of x will be equal to:

(a) $\angle DAC$ (b) $\angle BAD$

(c) $\angle ACE$ (d) $\angle EAC$

Ans. (A) (b) SAS Congruence Rule

Explanation: In ΔABC and ΔADE

$AB = AD$ [Given]

$\angle BAC = \angle DAE$ [Given]

$AC = AE$ [Given]

$\therefore \Delta ABC \cong \Delta ADE$ [By SAS Rule]

(B) *(c) $\angle BAC = \angle DAE$*

Explanation: In ΔABC and ΔADE

$AB = AD$ [Given]

$AC = AE$ [Given]

So, the required condition is $\angle BAC = \angle DAE$

(C) *(d) DE*

Explanation: $\because \Delta ABC \cong \Delta ADE$

[By SAS congruence rule]

$\therefore$ $BC = DE$ [By CPCT]

(D) *(d) $\angle EAC$*

Explanation: $\angle BAD = \angle EAC$ [Given]

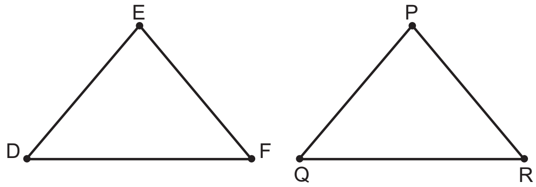
Add $\angle DAC$ on both sides, we get

$\angle BAD + \angle DAC = \angle EAC + \angle DAC$.

VERY SHORT ANSWER Type Questions (VSA)

[**1** mark]

6. If ΔPQR ≅ ΔEDF, then is it true to say that PR = EF? Give a reason for your answer.



[NCERT Exemplar]

Ans. Yes, if ΔPQR ≅ ΔEDF, then it means that corresponding angles and their sides are equal because we know that, two triangles are congruent, if the sides and angles of one triangle are equal to the corresponding sides and angles of other triangle.

∵ ΔPQR ≅ ΔEDF,

PQ = ED, QR = DF and PR = EF

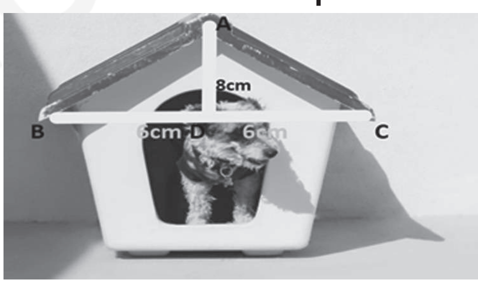
[These are corresponding sides]

Hence, it is true to say that PR = EF

SHORT ANSWER Type-I Questions (SA-I)

[**2** marks]

7. The front of a dog house has the dimensions (as shown in the figure). Prove that ΔADB ≅ ΔADC also find the perimeter of ΔABC.



Ans. In ΔABD and ΔACD,

∠ADB = ∠ADC	[Both 90°]
BD = DC = 6 cm	[Given]
AD = AD	[Common]

By SAS congruence rule,

ΔABD ≅ ΔACD

Now, AB = AC [By CPCT]

In ΔABD,

$$AB^2 = AD^2 + BD^2$$
$$AB^2 = (8)^2 + (6)^2$$
$$= 64 + 36$$
$$AB^2 = 100$$
$$AB = 10 \text{ cm}$$
$$AB = AC = 10 \text{ cm}$$

Therefore,

Perimeter of ABC = AB + AC + BC

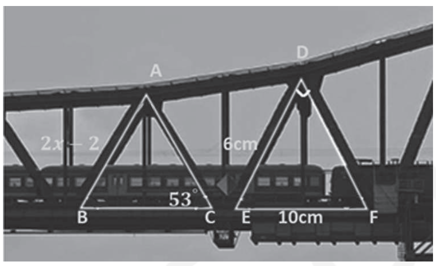
= (10 + 10 + 12) cm

= 32 cm

SHORT ANSWER Type-II Questions (SA-II)

[**3** marks]

8. If $\Delta ABC \cong \Delta DFE$, find the value of x.



Ans. Since, $\Delta ABC \cong \Delta DFE$ [Given]

$\therefore$ $\angle C = \angle E$

Since, $\angle C = 53^\circ$ [Given]

Therefore, $\angle E = 53^\circ$

In ΔDEF, $\angle D = 90^\circ$

$$EF^2 = DE^2 + DF^2$$
$$10^2 = 6^2 + DF^2$$
$$DF^2 = 100 - 36$$
$$DF^2 = 64$$
$$DF = \sqrt{64}$$
$$DF = 8 \text{ cm}$$

Also, $\angle C = \angle E$

$\therefore \qquad AB = DF \qquad [\Delta ABC \cong \Delta DFE]$

$2x - 2 = 8$

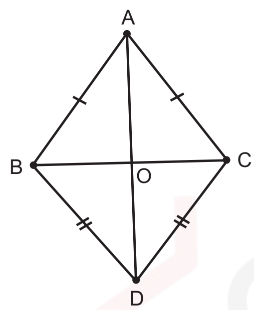
$2x = 10$

$x = 5$

LONG ANSWER Type Questions **(LA)**

[**4** & **5** marks]

9. ABC and DBC are two triangles on the same base BC such that A and D lie on the opposite sides of BC, AB = AC and DB = DC. Show that AD is the perpendicular bisector of BC.



[NCERT Exemplar]

Ans. Let AD intersect BC at O.

Then must prove that $\angle AOB = \angle AOC = 90^o$ and BO = OC

In ΔABD and ΔACD, we have

$AB = AC$ [Given]

$AD = DA$ [Common]

$BD = DC$ [Given]

$\therefore \quad \Delta ABD \cong \Delta ACD$ [By SSS]

$\Rightarrow \quad \angle BAD = \angle CAD$ [By CPCT]

In ΔABC,

$\angle AOB = \angle AOC$

[Angle opposite to equal sides are equal]

But $\quad \angle AOB + \angle AOC = 180^o$ [Linear pair]

$\Rightarrow \angle AOB + \angle AOB = 180^o$

$\Rightarrow \quad 2\angle AOB = 180^o$

$\therefore \quad \angle AOB = 90^o$

Hence, AD is perpendicular to BC and AD bisects BC.

$\therefore$ AD is a perpendicular bisector.